Two consecutive nondual Jacobian transfers

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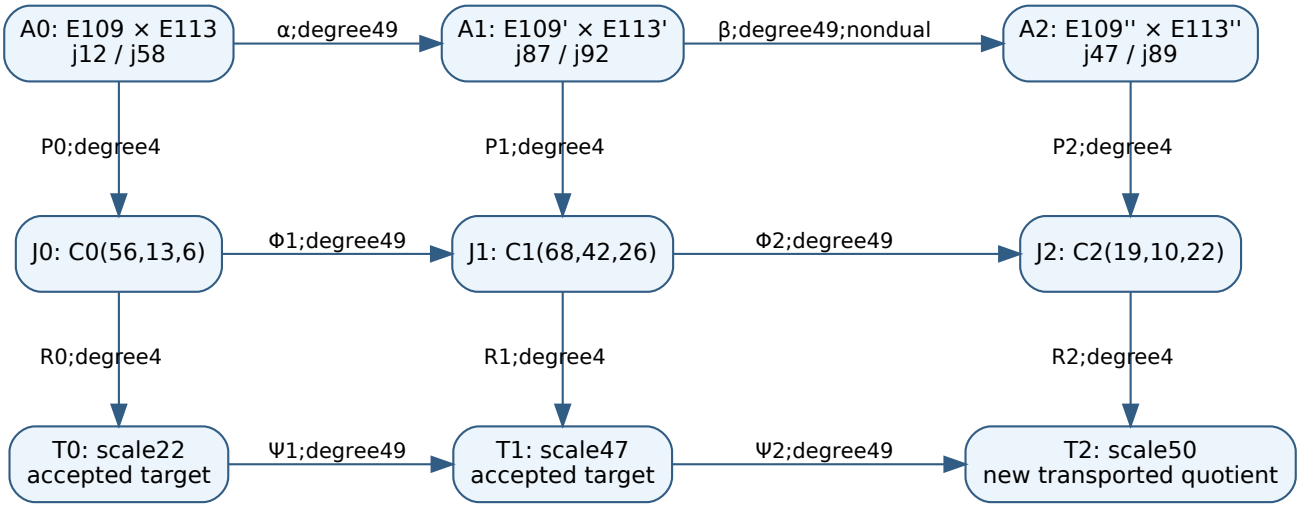
The second nondual seven-transfer Φ2 is implemented using the existing elliptic, quotient, finite-field and balanced-divisor engines. Both complete 12,317-class C1/C2 rational domains pass. Transporting the actual R1 kernel gives R2 and a regular (7,7) map Ψ2:T1→T2. Its written global construction and all109/113 original source controls are complete; the full consumer guard remains a separate checkpoint obligation. PCC6a2cfe/6e531a. [pcc:unverified]



Every displayed square commutes as an identity of global homomorphisms. Each Ψ has polarized-adjoint norm7. Their composite has degree2401 and adjoint norm49. All maps descend to F101. Kernel point enumeration has a different field requirement, described on the next page. [pcc:unverified]

Exact new target and independent controls

The C2 curve is $y^2=x^6+22x^4+10x^2+19$, with quotient charts (destination,shift,scale)=(1,41,18), (0,37,18). The actual transported R2 matching is (0,4),(1,5),(2,3), determinant49. Raw target coefficients are [49,99,36,35,23,80,76]; ordinate scale50 gives $T2=Jac(y^2=x^6+17x^5+92x^4+39x^3+43x^2+93x+95)$. [pcc:unverified]

The versioned independent branch oracle reuses the complete accepted source as a byte-identical prefix. All120 triple-induced PGL2 transformations are examined; only identity preserves the six actual T2 roots. Hyperelliptic theory and strong Torelli identify polarized target automorphisms as ±1. A nonzero odd rational image rules out the negative Frobenius twist. Read SUCCESSOR_CHAIN_PROOF.md. [pcc:unverified]

Proof, evidence and the remaining computational question

What the construction proves

R2 Φ2 R1Hat kills all16 geometric T1[2] classes. Finite-etale scheme kernel inclusion and the abelian quotient property give its regular factor Ψ2 through [2]. Degrees give49; cancellation in torsion-free Hom groups gives both dual norms7. This does not divide geometric two-torsion by an odd rational group order. [pcc:unverified]

For either original prime subgroup, the two-hop reverse route followed by the original norm is multiplication by196. Its inverses are104 modulo109 and64 modulo113. Target quotient coordinates equal4·β·α(P). Native tests check every109/113 scalar, both naive/quarter evaluators, nonzero full order and both commuting squares. These controls preserve the original DLP; they do not solve it. [pcc:unverified]

The Richelot kernels avoid both embedded elliptic factors. Their trace−7 factors have j87 and47, so rational trace-type uniqueness distinguishes T1 andT2 even as unpolarized F101 abelian varieties. The same argument distinguishes T0 andT2 usingj12 and47. No algebraic-closure distinction is claimed. [pcc:unverified]

Kernel fields and independent replay scope

Each individual seven-transfer has49 kernel points over the existing $F101^{6}$. The composite has group Z/49 × Z/49 and Frobenius33/27 modulo49, with orders42/14. Its full2401-point kernel needs $F101^{42}$; only49 points lie over $F101^{6}$. Prime-field maps do not require listing that full kernel. [pcc:unverified]

All49 new Ψ2 kernel points are exported. A separate flatFpR/polynomial replay checks their actual T1 divisor validity, distinctness, semantic basis and Frobenius5/6, with fixed counts1/7/1/49 at degrees1/2/3/6. This replay does not check independent Jacobian additions or kernel membership. Standalone six-test/strictClippy14 receipt: PCC78b6c4. [pcc:unverified]

Five TwoHopTransport Lean lemmas prove the conditional two-square, norm49/reverse196 and recovery implications. All five pass the kernel gate/statement bindings, with zero trusted native evaluation; the complete written source census is114. Fields, Picard functoriality, quotient descent, polarizations, Torelli and subvariety uniqueness remain written dependencies. PCCcbb605/78b6c4. Formalization is partial. [pcc:unverified]

Acceptance and next research

SUCCESSOR_CHECKPOINT.md records exact local/CI revisions and pending full consumer acceptance. Retained failures include missing-trait compilation03 and index-only-loop Clippy10. Sources: HLP §3.2, Kunzweiler §3.2, Milne AV8.12/10.20 and JV12.1; primary hashes are in the parent proofs. Locked atlas/KB and ledger-fork limits persist. [pcc:unverified]

There is no timing, speedup, cryptographic-size or ECDLP-algorithm advantage claim. S3 admission and current live budget are unavailable, so performance experiments remain blocked. Next evaluate the cost and usefulness of these transfers under an admitted matched protocol; meanwhile continue unblocked structural and implementation work. [pcc:unverified]